

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

● ANSWER KEY

Two-variable data: Models and scatterplots

03

10 answers · Rationale-rich · Teach from doc

Q1**D****D. 2010–2011**

Choice D is correct. The change in the number of 3-D movies released between any two consecutive years can be found by first estimating the number of 3-D movies released for each of the two years and then finding the positive difference between these two estimates. Between 2003 and 2004, this change is approximately $2 - 2 = 0$ movies; between 2008 and 2009, this change is approximately $20 - 8 = 12$ movies; between 2009 and 2010, this change is approximately $26 - 20 = 6$ movies; and between 2010 and 2011, this change is approximately $46 - 26 = 20$ movies. Therefore, of the pairs of consecutive years in the choices, the greatest increase in the number of 3-D movies released occurred during the time period between 2010 and 2011.

Choices A, B, and C are incorrect. Between 2010 and 2011, approximately 20 more 3-D movies were released. The change in the number of 3-D movies released between any of the other pairs of consecutive years is significantly smaller than 20.

Q2**B****B. 39°F**

Choice B is correct. The line of best fit passes through the point $(8,500, 39)$. Therefore, the line of best fit predicts a temperature of 39°F for a location in Lake Tahoe Basin with an elevation of 8,500 feet.

Choice A is incorrect. This is the lowest temperature listed on the scatterplot, and the line of best fit never crosses this value for any of the elevations shown. Choice C is incorrect. According to the line of best fit, the temperature of 41°F is predicted for an elevation of slightly greater than 7,500 feet, not an elevation of 8,500 feet. Choice D is incorrect. According to the line of best fit, the temperature of 43°F is predicted for an elevation of roughly 6,700 feet, not an elevation of 8,500 feet.

Q3**C****C.** 11

Choice C is correct. On the line of best fit, an x -value of 1,200 corresponds to a y -value between 10 and 12. Therefore, of the given choices, 11 is closest to the y -value predicted by the line of best fit at $x = 1,200$.

Choice A is incorrect. This is the integer value closest to the y -value predicted by the line of best fit at $x = 1,800$.

Choice B is incorrect. This is the integer value closest to the y -value predicted by the line of best fit at $x = 1,500$.

Choice D is incorrect. This is the integer value closest to the y -value predicted by the line of best fit at $x = 600$.

Q4**40**

The correct answer is 40. For each data point on the scatterplot, the x -value represents the growing season after transplantation and the y -value represents the weight, in pounds, of all nectarines that grew on the tree during the season. The scatterplot shows a data point at 4, 40. It follows that during the 4th growing season after the tree's transplantation, 40 pounds of nectarines grew on the tree.

Q5**B****B.** 35

Choice B is correct. In the given scatterplot, the x -values represent the distance above sea level, in feet, and the y -values represent the temperature, in $^{\circ}\text{F}$. The point on the line of best fit with an x -value of 4,000 has a corresponding y -value of 35. Therefore, at a distance of 4,000 feet above sea level, the temperature predicted by the line of best fit is 35°F .

Choice A is incorrect. This is the temperature, in $^{\circ}\text{F}$, predicted by the line of best fit at a distance of 0 feet above sea level.

Choice C is incorrect. This is the measured temperature, in $^{\circ}\text{F}$, at a distance of 6,000 feet above sea level.

Choice D is incorrect and may result from conceptual or calculation errors.

Q6**A****A.** Increasing linear

Choice A is correct. The graph of function f shows that as x increases, $f(x)$ also increases, which means $f(x)$ is an increasing function. The graph of f is a line, which indicates a constant rate of change. A function that has a constant rate of change is a linear function. Therefore, function f can be described as increasing linear.

Choice B is incorrect. For a decreasing function, as x increases, $f(x)$ decreases, rather than increases.

Choice C is incorrect. The graph of an exponential function isn't a line.

Choice D is incorrect. For a decreasing function, as x increases, $f(x)$ decreases, rather than increases, and the graph of an exponential function isn't a line.

Q7**A**

A. $y = x + 3.4$

Choice A is correct. The line of best fit shown has a positive slope and intersects the y -axis at a positive y -value. The graph of an equation of the form $y = mx + b$, where m and b are constants, has a slope of m and intersects the y -axis at a y -value of b . Of the given choices, only $y = x + 3.4$ represents a line that has a positive slope, 1, and intersects the y -axis at a positive y -value, 3.4.

Choice B is incorrect. This equation represents a line that intersects the y -axis at a negative y -value, not a positive y -value.

Choice C is incorrect. This equation represents a line that has a negative slope, not a positive slope.

Choice D is incorrect. This equation represents a line that has a negative slope, not a positive slope, and intersects the y -axis at a negative y -value, not a positive y -value.

Q8**C**

C. 1,300

Choice C is correct. The data in the scatterplot show an increasing linear trend. The density when the juice concentration is 60% will be between the densities shown at about 53% and 67% concentration, or between about 1,255 and 1,340 kg/m^3 . Of the choices given, only 1,300 falls within this range.

Choices A, B, and D are incorrect. These are the approximate densities of grape juice with a concentration of 45%, 55%, and 70%, respectively.

Q9**C****C.** 4.6

Choice C is correct. According to the line of best fit, a planetoid with a distance from the Sun of 1.2 AU has a predicted density between 4.5 g/cm^3 and 4.75 g/cm^3 . The only choice in this range is 4.6.

Choices A, B, and D are incorrect and may result from misreading the information in the scatterplot.

Q10**C****C.** From $x = 2$ to $x = 3$

Choice C is correct. The scatterplot shows that there was an increase in recorded temperature from $x = 2$ to $x = 3$ and from $x = 6$ to $x = 7$. When $x = 2$, the recorded temperature was approximately 60°F and when $x = 3$, the recorded temperature was greater than 70°F . This means that the increase in recorded temperature from $x = 2$ to $x = 3$ was greater than $(70 - 60)^\circ\text{F}$, or 10°F . When $x = 6$, the recorded temperature was greater than 60°F and when $x = 7$, the recorded temperature was less than 70°F . This means that the increase in recorded temperature from $x = 6$ to $x = 7$ was less than $(70 - 60)^\circ\text{F}$, or 10°F . It follows that the greatest increase in recorded temperature took place from $x = 2$ to $x = 3$.

Choice A is incorrect. The increase in recorded temperature from $x = 6$ to $x = 7$ was less than the increase in recorded temperature from $x = 2$ to $x = 3$.

Choice B is incorrect. From $x = 5$ to $x = 6$, a decrease, not an increase, in recorded temperature took place.

Choice D is incorrect. From $x = 1$ to $x = 2$, a decrease, not an increase, in recorded temperature took place.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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